FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE (Main), 2015

HALF COURSE TEST –VI

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 03 and 10 to 12)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 mark** for wrong answer.

Section-A (04 to 09 and 13 to 30) contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Single Correct Choice Type

*This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

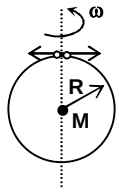
1. A cylinder released from the top of an inwclined plane of inclination θ and having coefficient of friction μ rolls without sliding and reaches the bottom with speed v_r. Another cylinder, released from the top of the same plane rolls with sliding till it reaches bottom with speed v_s. Then (given that $\mu > \frac{\tan\theta}{3}$)

   (A) $V_s > V_r$ (B) $V_r > V_s$
   (C) $V_r = V_s$ (D) none of these

2. A uniform rod of length ℓ is suspended from one end such that it is free to rotate about an axis passing through that end and perpendicular to the rod. A small sphere of same mass as that of rod strikes the other end of the rod and sticks to it. The minimum velocity of the sphere for which the rod can make full rotations is

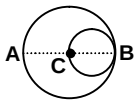
   (A) $\sqrt{8g\ell}$ (B) $\sqrt{4g\ell}$
   (C) $\sqrt{5g\ell}$ (D) none of these

3. From A ring of mass M and radius R rotating with angular velocity ω about it's diameter (being vertical) two beads of mass m are released from rest from the upper end of the diameter. The angular velocity of the ring when the two bead's meet at the lower end is



   (A) ω (B) 2ω
   (C) $M\omega/M+2m$ (D) none of these

4. The moment of inertia of a uniform disc with centre C and radius R, having circular cavity of diameter CB = R about an axis passing through A, B and C and perpendicular to the plane of disc is I_A, I_B and I_C respectively. Then



   (A) $I_C > I_B > I_A$ (B) $I_B > I_C > I_A$
   (C) $I_A > I_C > I_B$ (D) $I_B > I_A > I_C$

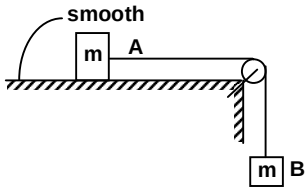
Space for rough work

5. If moment of inertia of a rod of mass m and length ℓ about an axis through it's end and perpendicular to the rod is I, then the moment of inertia about the same axis if the rod is bent at 90° at it's mid point is

(A) $\frac{5}{4}I$ (B) $\frac{5}{8}I$

(C) $\frac{13}{8}I$ (D) none of these

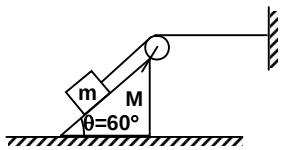
6. The system shown in figure is released from rest, if block B is stopped for a moment after 1 sec from the start, the time after which the string will again become taut is



(A) 1 sec (B) 2 sec

(C) 3 sec (D) none of these

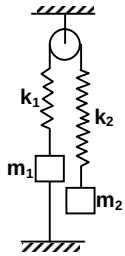
7. The system shown in figure is released from rest. The ratio of velocities of block and wedge after 2 sec from the start of motion is (Assuming all the surface to be frictionless)



(A) 1/2 (B) 1

(C) 2 (D) none of these

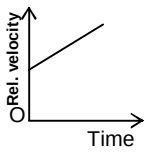
8. In the figure shown pulley string and spring are massless and the system is in equilibrium. If the string connected to the ground is cut, the instantaneous acceleration of m_1 mass is



(A) g/2 (B) g/3

(C) $\frac{m_1 g}{m_2}$ (D) $\frac{(m_2 - m_1)g}{m_1}$

9. A particle is observed from two frames S_1 and S_2. The graph of relative velocity of S_1 with respect to S_2 is shown in figure. Let F_1 and F_2 be the pseudo forces on the particle when seen from S_1 and S_2 respectively. Which of the following is impossible?

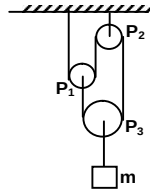


(A) $F_1 = 0, F_2 \neq 0$ (B) $F_1 \neq 0, F_2 = 0$

(C) $F_1 \neq 0, F_2 \neq 0$ (D) $F_1 = 0, F_2 = 0$

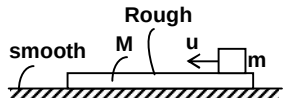
Space for rough work

10. In the figure shown neglect the masses of the pulley and also neglect friction. Then



(A) tension in the string connected to block is mg/2
(B) Acceleration of pulley P_1 is g upwards
(C) Acceleration of pulley P_3 is g/2 downwards
(D) Acceleration of block is g downwards.

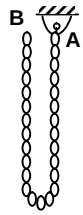
11. A block of mass m is given velocity u over a long plank of mass M as shown in figure. The velocity of the block after a long time is



(A) $\frac{mu}{M}$ (B) $\frac{mu}{M+m}$

(C) $\frac{u}{2}$ (D) $\frac{Mu}{M+m}$

12. The end B of the chain of mass per unit length λ and length ℓ is released from rest as shown in figure. The force at the hinge when the end B is at $\ell/4$ from the ceiling is



(A) $\frac{3}{4}\lambda gL$ (B) $\frac{\lambda gL}{4}$

(C) $\frac{4}{3}\lambda gL$ (D) none of these

13. Choose the correct statement.
(A) Work done by friction force is always negative.
(B) Static friction is always greater than kinetic friction
(C) When the body slides on a surface the value of friction is equal to limiting friction.
(D) The coefficient of friction between two bodies does not depend upon the area of contact.

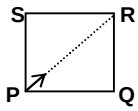
Space for rough work

14. For a body moving on a circular path, the angular velocity, linear velocity, angular acceleration, centripetal acceleration and tangential acceleration of the particle at any instant are $\vec{\omega}, \vec{v}, \vec{\alpha}, \vec{a}_c$ and $\vec{a}_t$ respectively. Which of the following relation is incorrect.

(A) $\vec{\omega} \perp \vec{v}$ (B) $\vec{a}_t \perp \vec{a}_c$

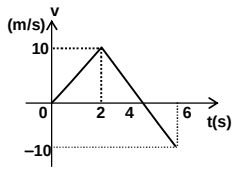
(C) $\vec{\omega} \perp \vec{a}_t$ (D) $\vec{\alpha} \perp \vec{\omega}$

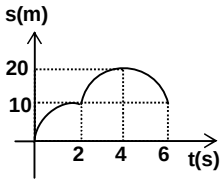
15. A particle is projected from corner P of an elevator moving downwards with acceleration a towards diagonally opposite corner R. Then

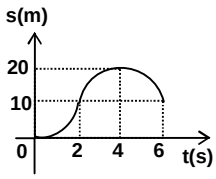


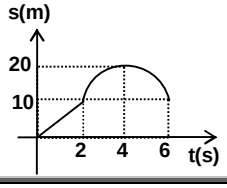
(A) particle will hit R only when a = g.
(B) particle will hit the wall QR if a > g
(C) particle will hit the roof RS if a < g.
(D) data insufficient

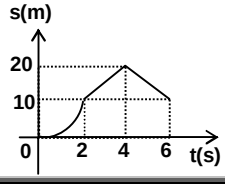
16. For the given velocity time curve, displacement is zero at t = 0 the corresponding displacement time curve is



(A)


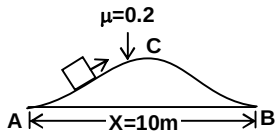
(B)


(C)


(D)


Space for rough work

17. A block of mass 1 kg is pulled along the curve path ACB by a tangential force as shown in figure. The work done by the friction force when the block moves from A to B is



(A) 5 J (B) 10 J
(C) 20 J (D) none of these

18. An inclined plane of height 10 m and angle of inclination 30° with the horizontal is having a groove at an angle of 45° with the line of greatest slope. A particle is released from rest from the top of incline plane in the groove, the time taken by the particle to reach the bottom of the groove is ($g = 10\ m/s^2$)

(A) 1 sec (B) 2 sec
(C) 4 sec (D) none of these

19. A light rod of length L, is hanging from on the vertical wall of a vehicle moving with acceleration $\sqrt{3}$ g having a small mass attached at it's one end is free to rotate about an axis passing through the other end. The minimum velocity given to the mass at it's equilibrium position so that it can complete vertical circular motion is

(A) $\sqrt{5gL}$ (B) $\sqrt{4gL}$
(C) $\sqrt{8gL}$ (D) none of these

20. A particle moves with a speed $v = |t - 3|$ m/s along a straight line, where t is time in seconds. Distance travelled by the particle during first 5 seconds is equal to:

(A) 1.5 m (B) 2m
(C) 6.5 m (D) none of these

21. Two particles A and B starts from rest and move for equal time on a straight line. Particle A has an acceleration of z m/s^2 for the first half of the total time and 2zm/s^2 for the second half. The particle B has an acceleration of 2z m/s^2 for the first half of the total time and z m/s^2 for the second half. Which particle has covered larger distance?

(A) A has covered the larger distance (B) B has covered the larger distance
(C) both have covered the same distance (D) none of these

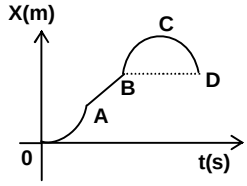
Space for rough work

22. A uniform circular ring of mass m and radius R is rotating in a gravity free space with angular velocity ω about its own axis. Tension in the ring is

(A) $\frac{mR\omega^2}{2\pi}$ (B) $\frac{mR\omega^2}{4\pi}$

(C) zero (D) $mR\omega^2$

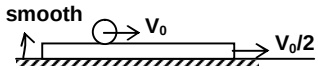
23. The displacement time graph for a particle in motion is shown in figure. For which portion of the graph the force acting on the particle is opposite to the direction of velocity



(A) AB (B) OA

(C) BC (D) CD

24. A cylinder of mass m and radius R is given velocity V_0 over a plank of same mass having velocity $V_0/2$. The velocity of the cylinder when it starts rolling without slipping is



(A) $\frac{V_0}{8}$ (B) $\frac{7V_0}{8}$

(C) $\frac{V_0}{4}$ (D) $\frac{3V_0}{4}$

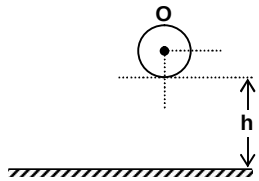
25. In the previous question if coefficient of friction between the plank and cylinder is μ, the work done by friction in the motion is

(A) $-\frac{1}{16}mv_0^2$ (B) $-\frac{1}{32}\mu mg$

(C) $-\frac{1}{32}mv_0^2$ (D) zero

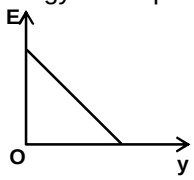
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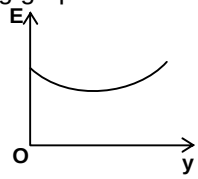
26. A small sphere and a big sphere are released from rest with a very small gap from height h as shown in figure. The mass of bigger sphere is very large as compared to mass of smaller sphere the height from the point of collision of smaller sphere with the bigger sphere to which the smaller sphere will rise if all the collision is elastic

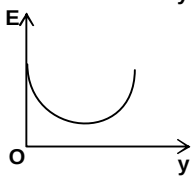


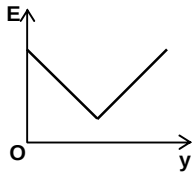
(A) 2h (B) 4h
(C) 6h (D) 9h

27. A particle is projected from a point at an angle with the horizontal at, t = 0. At an instant t, if p is magnitude of linear momentum, x is horizontal displacement, y is vertical displacement E is kinetic energy of the particle, then which of the following graph is correct.

(A) 

(B) 

(C) 

(D) 

28. A particle on the top of a fixed sphere of radius R is given horizontal velocity $\sqrt{2gR}$, the angle measured from vertical at which it will leave contact with the sphere is

(A) zero (B) $\cos^{-1}(2/3)$
(C) 30° (D) 45°

29. Work will be done by a force on an object, if
(A) the object is stationary but the point of application of the force moves.
(B) the force is always perpendicular to its velocity
(C) the object moves in such a way that the force is perpendicular to its displacement.
(D) none of these

30. A block is suspended by an ideal spring constant K. If the block is pulled down by constant force F and if maximum displacement of block from it's initial position of rest is z, then
(A) z = F/K
(B) z = 2F/K
(C) work done by force F is equal to 2Fz.
(D) increase in potential energy of the spring is $\frac{1}{2}Kz^2$

Space for rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

*This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

1. How many substituted products are possible on monochlorination of isohexane (including isomers)?
   (A) 5 (B) 6
   (C) 7 (D) 8

2. For the reaction: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ equilibrium constant is
   (A) $\frac{(\eta_{NH_3})^2}{(\eta_{N_2})(\eta_{H_2})^3} V$ (B) $\frac{(\eta_{NH_3})^2}{(\eta_{N_2})(\eta_{H_2})^3} V^2$
   (C) $\frac{(\eta_{NH_3})^2}{(\eta_{N_2})(\eta_{H_2})^3} V^3$ (D) $\frac{(\eta_{NH_3})^2}{(\eta_{N_2})(\eta_{H_2})^3}$

3. The rate constant for the reaction: $2N_2O_5 \longrightarrow 4NO_2 + O_2$ is 6×10^{-5} s^{-1}. If the rate is 2.40×10^{-5} mol $L^{-1}s^{-1}$, then the concentration of N_2O_5 (in mol L^{-1}) is
   (A) 0.8 (B) 0.4
   (C) 0.2 (D) 0.1

4. Dolomite is mineral whose formula is
   (A) $CaCO_3$ (B) $CaCO_3.MgCO_3$
   (C) $MgCO_3$ (D) $CaSO_4.2H_2O$

5. Which of the following thermodynamic relation is correct for ideal gas?
   (A) $\left(\frac{\partial V}{\partial T}\right)_P = 0$ (B) $\left(\frac{\partial P}{\partial T}\right)_V = 0$
   (C) $\left(\frac{\partial E}{\partial V}\right)_T = 0$ (D) $\left(\frac{\partial V}{\partial T}\right)_P = 0$

Space for rough work

6. How many bonds are there in naphthalene?
(A) 18σ, 4π (B) 18 σ, 5π
(C) 19 σ, 5π (D) 19 σ, 4π

7. Which of the following is correct relation between mole fraction of solute into molarity of solution?
(A) $M = \frac{x_2}{M_2\left(X_1M_1 + x_2M_2\right)}$ (B) $M = \frac{\left(x_1M_1 + x_2M_2\right)}{x_2M_2}$
(C) $M = \frac{x_2\rho}{\left(x_1M_1 + x_2M_2\right)}$ (D) $M = \frac{\left(1 - m\rho\right)}{mM_2}$

8. Which of the following expression is correct for rate of diffusion of 2 gases?
(A) $r_{H_2} = r_{D_2}$ (B) $r_{CO_2} = r_{NO_2}$
(C) $r_{N_2} = r_{CO_2}$ (D) $r_{CO_2} = r_{N_2O}$

9. K_{sp} of a substance MX_2 is 4×10^{-6} mol^3L^{-3}. Molecular mass of the substance is 200 its solubility would be
(A) 0.2 g/L^{-1} (B) 2 g/L
(C) 20 g/L (D) 200 g/L

10. Which of the following represents the given mole of hybridization sp – sp – sp^2 – sp^2 – sp^3 from left to right?
(A) $CH_3 - CH_3 - CH_2 = CH - NH_2$ (B) $CH \equiv CH - CH_2 - CH_2 - NH_2$
(C) $CH \equiv CH - CH_2 - NH - CH_3$ (D) $CH \equiv CH - N = N - CH_3$

11. The hybridization of diborane is
(A) sp (B) sp^2
(C) sp^3 (D) sp^3d

12. Which of the following ion has the maximum value of magnetic moment?
(A) Mn^{2+} (B) Fe^{2+}
(C) Co^{2+} (D) Ni^{2+}

Space for rough work

13. 2 moles of SO_2 and 1 mole of O_2 are allowed to react in a container of capacity 2 litre and following equilibrium is allowed to establish: $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

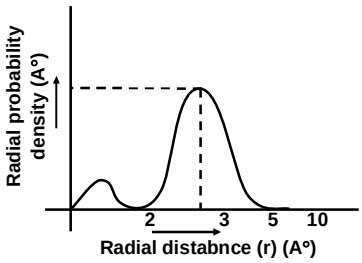
The equilibrium mixture requires 200 ml of 0.1 M acidified $K_2Cr_2O_7$ solution. Hence K_c for the above equilibrium is approximately

(A) 6.96×10^2 (B) 6.96×10^3

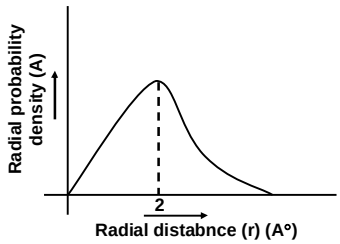
(C) 6.96×10^4 (D) 6.96×10^5

14. Which of the following radial probability curve is correct for a 3p – orbital? (Assume that all graphs have been plotted for the same parameters. In all these curves principal quantum number < 4)

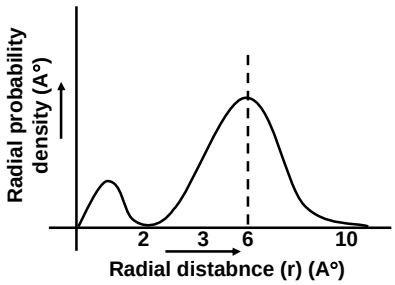
(A)



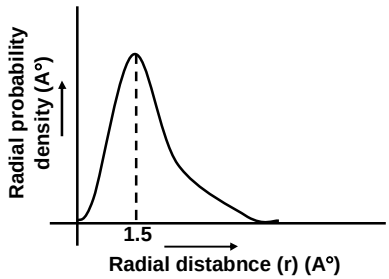
(B)



(C)



(D)



15. Which of the following species is incorrect method with its shape?

(A) $ClF_3 \longrightarrow$ T – shaped (B) $SO_3^{2-} \longrightarrow$ tetrahedral

(C) $C_2O_4^{2-} \longrightarrow$ planar (D) $I_3^- \longrightarrow$ linear

Space for rough work

16. Identify the incorrect statement (s) among the following:
(A) if spin quantum number curve having 3 possible values then 4th period of periodic table would have maximum of 27 elements.
(B) platinum (Pt) is one of the typical transition metal.
(C) total number of f – electrons in Gd (at. No. 64) is 8.
(D) atomic radius of Ga is nearly equal to that of Al.

17. To prepare a buffer of pH 8.26 amount of $(NH_4)_2SO_4$ to be added into 500 ml of 0.01 M NH_4OH solution [$pk_a(NH_4^+) = 9.26$] is
(A) 0.05 mole (B) 0.025 mole
(C) 0.10 mole (D) 0.005 mole

18. Consider the following equilibria:

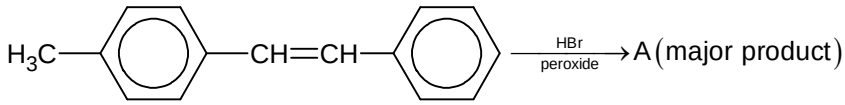
$$CH_3COOH + H_2O \rightleftharpoons H_3O^+ + CH_3COO^-$$

$$H_2SO_4 + CH_3COOH \rightleftharpoons CH_3COOH_2^+ + HSO_4^-$$

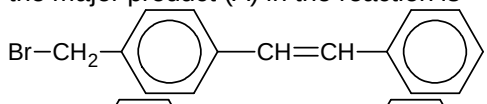
$$H_2SO_4 + HNO_3 \rightleftharpoons [H_2NO_3]^+ + HSO_4^-$$

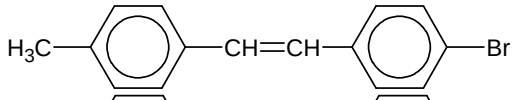
$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

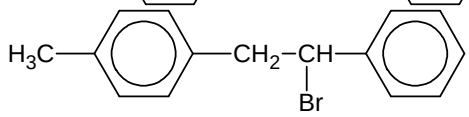
and, now select the incorrect statement among the following
(A) H_2SO_4 is a stronger acid than HNO_3
(B) NH_4^+ is a weaker acid than H_3O^+
(C) acetic acid is stronger acid than H_2O
(D) H_2O is a weaker acid than H_2SO_4 and HNO_3 both

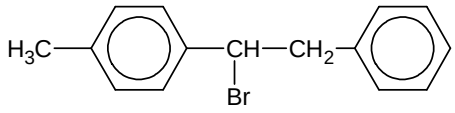
19. 

Hence the major product (A) in the reaction is

(A) 

(B) 

(C) 

(D) 

Space for rough work

20. An inorganic compound (X) on hydrolysis produces a gas which on treatment with sodium followed by its reaction with ethyl chloride gives another compound (Y). Compound (Y) on heating with P – 2 catalyst gives (2Z) – pent – 2 – ene as major product. Hence the inorganic compound (X) is
(A) Tl_4C (B) BaC_2
(C) SiC (D) Mg_2C_3

21. Which of the following metal reacts most vigorously with water?
(A) Li (B) Na
(C) K (D) Be

22. Boron trichloride on heating with ammonium chloride at 140°C gives a compound (A) which on reduction with sodium borohydride followed by hydrolysis gives another compound (B). Hence compound (A) and (B) are respectively.
(A) inorganic graphite & ammonia (B) inorganic benzene & orthoboric acid
(C) trichloroborazine & orthoboric acid (D) diborane & orthoboric acid

23. (S) – 2 – chloropentane on reaction with Cl_2 (g) in the presence of U.V. light, gives 'N' number of optically active dichloro derivatives. Hence the value of N is
(A) 5 (B) 6
(C) 7 (D) 4

24. Which of the following statement regarding keto - enol forms of acetone and ethyl actoacetate is correct?
(A) both are more stable in enol form than in keto form
(B) both are more stable in keto form than in enol form
(C) acetone is more stable in keto form whereas enol form is more stable in ethyl acetoacetate
(D) acetone is more stable in enol form whereas keto form is more stable in ethyl actacetate

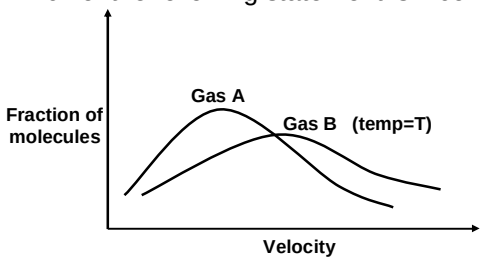
25. 3 mole of KIO_3, 4 mole of KI and 5 mole of HCl are allowed to react according to the equation given below:

$$KIO_3 + 2KI + 6HCl \longrightarrow 3ICl + 3KCl + 3H_2O$$

The number of moles of products are
(A) 27 (B) 18
(C) 7.5 (D) 9.5

Space for rough work

26. Which of the following statement is incorrect for the given graph?

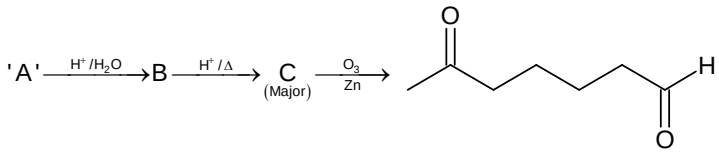


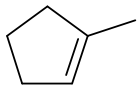
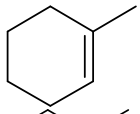
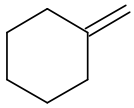
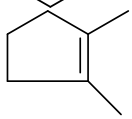
(A) The molecular weight of gas A will be greater than gas B
(B) The average kinetic energy of both the gases will be similar at a particular temperature
(C) The average velocity of particles of gas A will always be less than of particles of gas B at temperature T.
(D) The velocity of any particle of gas A can not be greater than velocity of any particle of gas B at same temperature.

27. A water sample is found to have 384 ppm of SO_4^{2-}. What will be the degree of hardness of water sample?
(A) 200 ppm (B) 300 ppm
(C) 400 ppm (D) 100 ppm

28. How many structural isomers are possible with molecular formula C_7H_{16}?
(A) 6 (B) 8
(C) 10 (D) 12

29. Which of the following is mismatched?
(A) $LiCO_3 \Rightarrow$ least stable carbonate among alkali metal carbonates.
(B) $LiF \Rightarrow$ almost insoluble in water due to its high covalent character.
(C) $Ba(OH)_2 \Rightarrow$ most soluble hydroxide among alkaline earth metal hydroxide.
(D) $BeSO_4 \Rightarrow$ more soluble than $BaSO_4$ in water.

30. 

(A) and (C) are isomers.
Which of the following can be structure of (A)?
(A) 
(B) 
(C) 
(D) 

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

*This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.*

1. If the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is touched by $y = x$ at P such that $OP = 6\sqrt{2}$, where O is the origin, then the value of c is
   (A) 36 (B) 144
   (C) 72 (D) none of these

2. If α is the nth root of unity, then $1 + 2\alpha + 3\alpha^2 + ...$ to n terms is equal to
   (A) $-\frac{n}{(1-\alpha)^2}$ (B) $-\frac{n}{1-\alpha}$
   (C) $-\frac{2n}{1-\alpha}$ (D) $-\frac{2n}{(1-\alpha)^2}$

3. If $(3 + x^{2008} + x^{2009})^{2010} = a_0 + a_1x + a_2x^2 + ... + a_nx^n$, then the value of $a_0 - \frac{1}{2}a_1 - \frac{1}{2}a_2 + a_3 - \frac{1}{2}a_4 - \frac{1}{2}a_5 + a_6 - .....$ is
   (A) 3^{2010} (B) 1
   (C) 2^{2010} (D) none of these

4. Total number of words that can be formed using all letters of the word 'BRIJESH' that neither begins with 'I' nor ends with 'B' is equal to
   (A) 3720 (B) 4920
   (C) 3600 (D) 4800

5. If $\cos^2 A + \cos^2 B + \cos^2 C = 1$, then ΔABC is
   (A) equilateral (B) isosceles
   (C) right angled (D) none of these

Space for rough work

6. In triangle ABC, $\angle A = \frac{\pi}{2}$, then $\tan\left(\frac{C}{2}\right)$ is equal to

(A) $\frac{a-c}{2b}$ (B) $\frac{a-b}{2c}$

(C) $\frac{a-c}{b}$ (D) $\frac{a-b}{c}$

7. For the equation $\cos^{-1} x + \cos^{-1} 2x + \pi = 0$, the number of real solution is

(A) 1 (B) 2

(C) 0 (D) ∞

8. Let A $\equiv$ (3, –4), B $\equiv$ (1, 2), let P $\equiv$ (2k – 1, 2k + 1) be a variable point such that PA + PB is minimum. Then k is

(A) $\frac{7}{9}$ (B) 0

(C) $\frac{7}{8}$ (D) none of these

9. If x, 2y, 3z are in A.P., where the distinct numbers x, y, z are in G.P., then the common ratio of the G.P. is

(A) 3 (B) $\frac{1}{3}$

(C) 2 (D) $\frac{1}{2}$

10. If $|z^2 - 3| = 3|z|$ then the maximum value of $|z|$ is

(A) 1 (B) $\frac{3+\sqrt{21}}{2}$

(C) $\frac{\sqrt{21}-3}{2}$ (D) none of these

Space for rough work

11. Total number of values of a so that $x^2 - x - a = 0$ has integral roots, where $a \in N$ and $6 \le a \le 100$, is equal to
(A) 2 (B) 4
(C) 6 (D) 8

12. If normals are drawn from a point P(h, k) to the parabola $y^2 = 4ax$, then the sum of the intercepts which the normals cut off from the axis of the parabola is
(A) h + a (B) 3(h + a)
(C) 2(h + a) (D) none of these

13. The eccentric angle of a point on the ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$ at a distance of $\frac{5}{4}$ units from the focus on the positive x–axis, is
(A) $\cos^{-1}\left(\frac{3}{4}\right)$ (B) $\pi - \cos^{-1}\left(\frac{3}{4}\right)$
(C) $\pi + \cos^{-1}\left(\frac{3}{4}\right)$ (D) none of these

14. The eccentricity of the conic represented by $x^2 - y^2 - 4x + 4y + 16 = 0$ is
(A) 1 (B) $\sqrt{2}$
(C) 2 (D) $\frac{1}{2}$

15. If centre of a regular hexagon is at origin and one of the vertices on Argand diagram is 1 + 2i, then its perimeter is
(A) $2\sqrt{5}$ (B) $6\sqrt{2}$
(C) $4\sqrt{5}$ (D) $6\sqrt{5}$

16. The number of even divisors of the number N = 12600 is
(A) 72 (B) 54
(C) 18 (D) none of these

17. If x, y, z are in G.P. and $a^x = b^y = c^z$, then
(A) $\log_b a = \log_a c$ (B) $\log_c b = \log_a c$
(C) $\log_b a = \log_c b$ (D) none of these

Space for rough work

18. Consider the ten numbers ar, ar^2, ar^3, ..., ar^{10}. If their sum is 18 and the sum of their reciprocals is 6 then the product of these ten numbers, is
(A) 81 (B) 243
(C) 343 (D) 324

19. The value of m for which one of the roots of $x^2 - 3x + 2m = 0$ is double of one of the roots of $x^2 - x + m = 0$ is
(A) –2 (B) 1
(C) 2 (D) none of these

20. $x^2 - xy + y^2 - 4x - 4y + 16 = 0$ represents
(A) a point (B) a circle
(C) a pair of straight lines (D) none of these

21. The value of $\frac{1+2\log_3 2}{(1+\log_3 2)^2} + (\log_6 2)^2$ is
(A) 2 (B) 3
(C) 4 (D) 1

22. $\frac{\sin 3\theta + \sin 5\theta + \sin 7\theta + \sin 9\theta}{\cos 3\theta + \cos 5\theta + \cos 7\theta + \cos 9\theta}$ is equal to
(A) $\tan 3\theta$ (B) $\cot 3\theta$
(C) $\tan 6\theta$ (D) $\cot 6\theta$

23. If in a triangle, $\left(1-\frac{r_1}{r_2}\right)\left(1-\frac{r_1}{r_3}\right) = 2$, then the triangle is
(A) right angled (B) isosceles
(C) equilateral (D) none of these

24. $\tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1}\left(\frac{x-y}{x+y}\right)$ is
(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$
(C) $\frac{\pi}{4}$ (D) $\frac{\pi}{4}$ or $-\frac{3\pi}{4}$

Space for rough work

25. The circle $x^2 + y^2 + 2\lambda x = 0$, $\lambda \in R$, touches the parabola $y^2 = 4x$ externally. Then
(A) $\lambda > 0$ (B) $\lambda < 0$
(C) $\lambda > 1$ (D) none of these

26. In a triangle ABC, $A \equiv (\alpha, \beta)$, $B \equiv (2, 3)$ and $C \equiv (1, 3)$ and point A lies on line $y = 2x + 3$ where $\alpha \in I$. Area of ΔABC, Δ, is such that $[\Delta] = 5$. Possible coordinates of A are (where [.] denotes the greatest integer function)
(A) (2, 3) (B) (5, 13)
(C) (–5, –7) (D) (–3, –5)

27. Sum of common roots of the equations $z^3 + 2z^2 + 2z + 1 = 0$ and $z^{1985} + z^{100} + 1 = 0$ is
(A) –1 (B) 1
(C) 0 (D) none of these

28. The last two digits of the number 3^{400} are
(A) 81 (B) 43
(C) 29 (D) 01

29. If t_n denotes the nth term of the series $2 + 3 + 6 + 11 + 18 + \ldots\ldots$ then t_{50} is
(A) $49^2 - 1$ (B) 49^2
(C) $50^2 + 1$ (D) $49^2 + 2$

30. Number of ways in which Rs. 18 can be distributed amongst four persons such that no body receives less than Rs. 3 is
(A) 126 (B) 84
(C) 4! (D) none of these

Space for rough work